Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

Computer Graphics and Animation

BCA — Semester V — 2021 — Answer Sheet

Subject Code: CACS305

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

2. What do you mean by area filling? Explain boundary fill algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Define scaling transformation? Prove that two successive scaling are multiplicative.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. What is symmetric property of circle? Digitize a circle whose center is at origin and radius is 9.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Define window and view port. Explain 2D viewing transformation pipeline.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Reflect a prism A(0,0,0), B(1,1,0), C(1,2,2) and (0,2,0) about yz-plane which has been rotated previously with +90 degree about y-axis.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What do you mean by visible surface detection? Explain Z. buffer algorithm for visible surface detection.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What is animation? Explain the steps to create animation sequence.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Group C

Attempt any TWO questions[2x10=20]

9. Define refresh rate and persistence of monitor. Explain architecture of random scan display system with block diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. How can you draw line using Bradenham's line drawing algorithm? Trace it to rasterize the line with end points P(1, 4) and Q(7, 8).

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Explain Cohen Sutherland line clipping algorithm. Let ABCD be the rectangular window with A(20,20), B(90,20), C(90,70) and D(20,70). Use Cohen Sutherland algorithm to find the region codes and ellipse the lines with end point P(10,30) and Q(80,80)

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]
